

CACS101: Computer Fundamentals and Applications

Unit 3 – Operating System

PRAJWAL BISTA

Introduction to Operating System

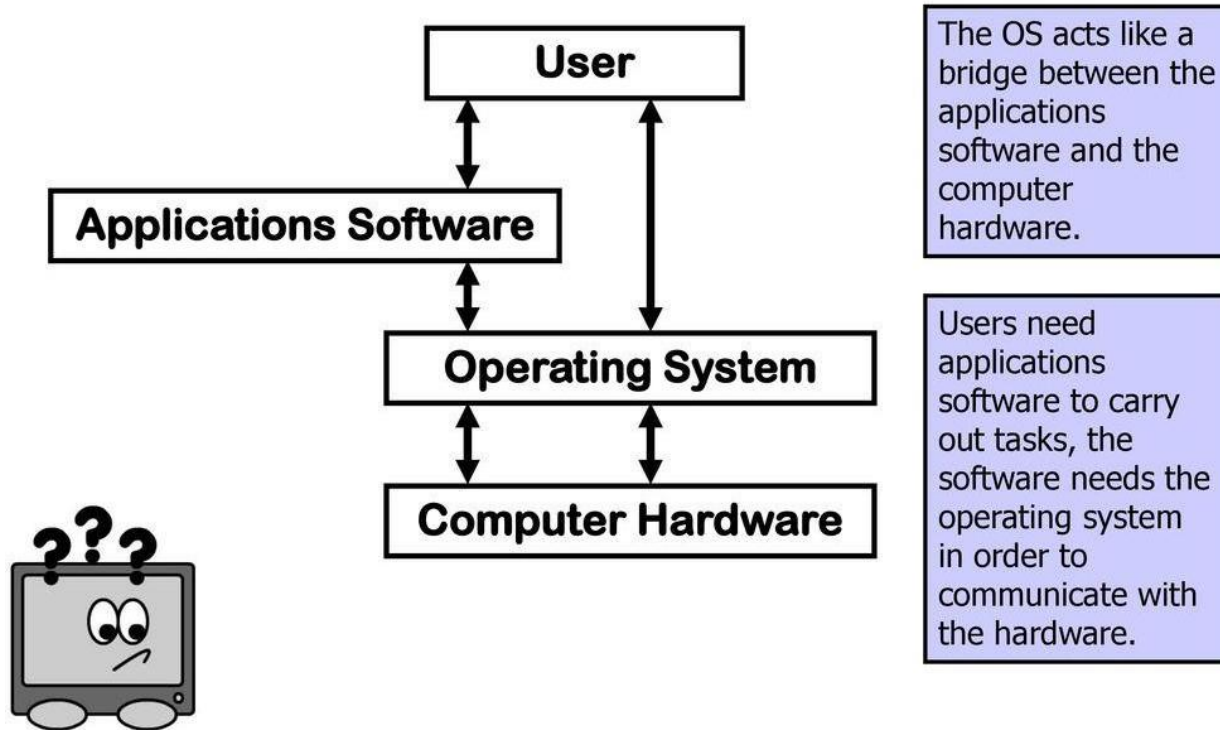
- An operating system is a system software which is set of specialized programs that are used to **control the resources** of a computer system.
- It is an organized set or collection of software that **controls the overall operation** of the computer system.
- Example: MS-DOS, Windows XP, UNIX, LINUX

Contd.

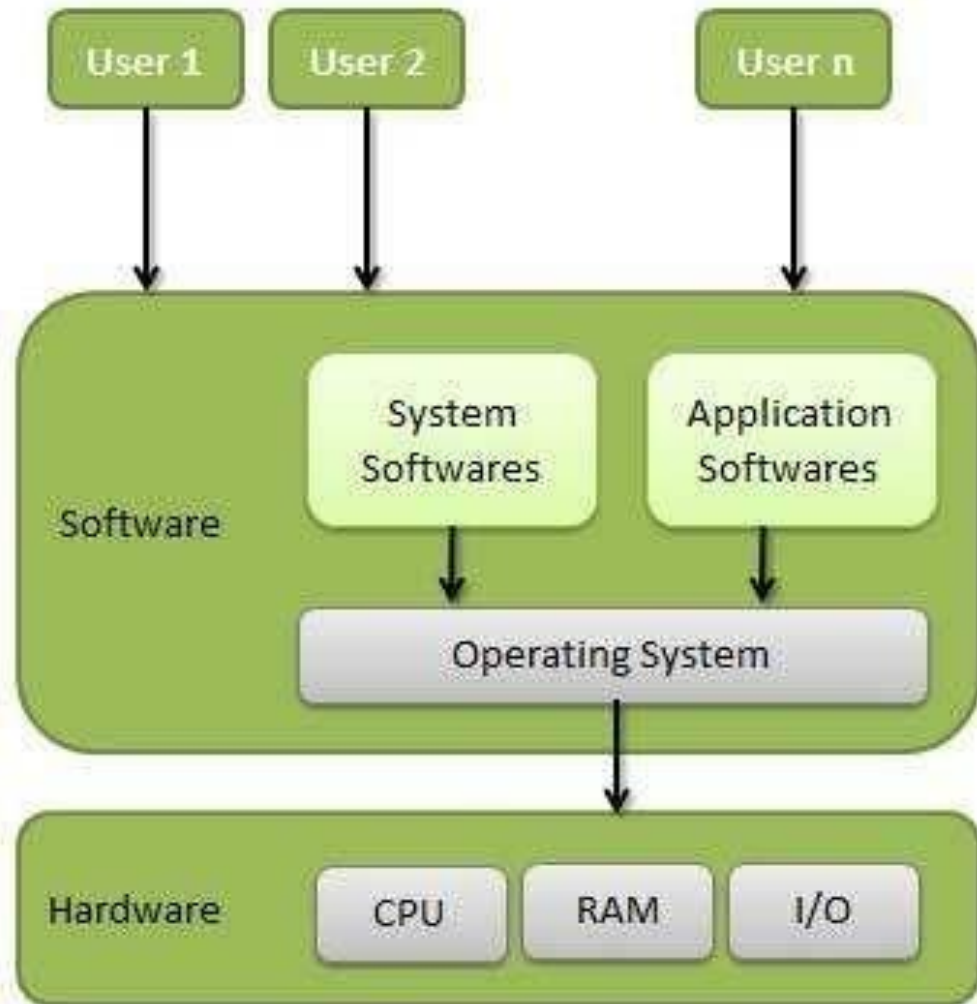
- An operating system is a program that acts as an intermediary between a user of a computer and the computer hardware.
- The purpose of an operating system is to provide an environment in which a user can execute programs. The primary goal of an operating system is thus to make the computer system convenient to use.
- A secondary goal is to use the computer hardware in an efficient manner.

Contd.

Operating Systems



Contd.



Functions of Operating System

- Input-Output Management
- File Management
- Memory Management
- Process Management
- Device Management
- Protection and Security
- User Interface or Command Interpreter

Input-Output Management

- Computer system includes the wide variety of input output devices such as printer, mouse, keyboard, speaker, light pen, etc.
- The speed, functions and designs of such devices are different from one another.
- The operating system handles such different I/O devices.

File Management

- File is a collection of data and information which is stored in computer system.
- The file management tasks include:
 - Create and delete both files and directories,
 - Provide access to files,
 - Allocate space for files,
 - Keep backup of files, and
 - Secure files.

Memory Management

- The task of memory management is to handle the allocation of memory to different processes and on completion of process execution, the memory is deallocated and made available to another process.
- Additionally, different processes that have been allocated memory should not interfere into each other's memory space. This requires some memory protection and sharing mechanism.
- Memory allocation, de-allocation, reallocation of free memory, and memory protection & sharing are the jobs of OS and is called memory management.

Process Management

- A process is a program in a state of execution. It is a unit of work for the operating system.
- A process can be created, executed, and stopped.
- The process management activities handled by the OS are:
 - Control access to shared resources like file, memory, I/O and CPU,
 - Control execution of applications
 - Create, execute and delete a process (system process or user process),
 - Cancel or resume a process
 - Schedule a process, and
 - Synchronization, communication and deadlock handling for processes.

Device Management

- OS manages and controls the devices attached to the computer. It handles the devices by combining both hardware and software techniques.
- The device management tasks handled by OS are:
 - Open, close and write device drivers, and
 - Communicate, control and monitor the device driver.

Protection and Security

- OS protects the resources of system.
- Following security measures are used to protect user data:
 - Protection against unauthorized access through login.
 - Protection against intrusion by keeping Firewall active.
 - Protecting the system memory against malicious access.
 - Displaying messages related to system vulnerabilities.

User Interface or Command Interpreter

- Operating system provides an interface between the computer user and the computer hardware.
- The user interface is a set of commands or a graphical user interface via which the user interacts with the applications and the hardware.

Functions of OS



Types of Operating System

- **Based on mode of user**
 - Single-user OS
 - Multiuser OS
- **Based on user interface**
 - Graphical user interface (GUI)
 - Character user interface (CUI)
- **Based on processing**
 - Batch OS
 - Multitasking OS
 - Distributed OS
 - Network OS
 - Real-Time OS

Single user OS

- **Single User/Single Tasking OS**
 - An operating system that allows a single user to perform only one task at a time is called a Single-User Single-Tasking Operating System.
 - Functions like printing a document, downloading images, etc., can be performed only one at a time.
 - This operating system occupies less space in memory.
 - Examples include MS-DOS, Palm OS, etc.

Single user OS

- **Single User/Multi Tasking OS**
 - An operating system that allows a single user to perform more than one task at a time is called Single-User Multitasking Operating System.
 - Functions like you can write any text while surfing the internet, downloading images, watching movies, etc., can be performed at the same time.
 - It is time saving as it performs multiple tasks at a time yielding high productivity.
 - Examples include Microsoft Windows and Macintosh OS.

Multuser OS

- Multi-user operating system allows concurrent access by multiple users of a computer.
- It allows many different users to take advantage of the computer resources simultaneously.
- It is often used in business and offices where different users need to access the same resources.

Graphical User Interface

- GUI stands for Graphical User Interface.
- This is a type of user interface where user interacts with the computer using graphics.
- Graphics include icons, navigation bars, images etc. Mouse can be used while using this interface to interact with the graphics.
- It is a very user-friendly interface and requires no expertise.
Eg: Windows has GUI.

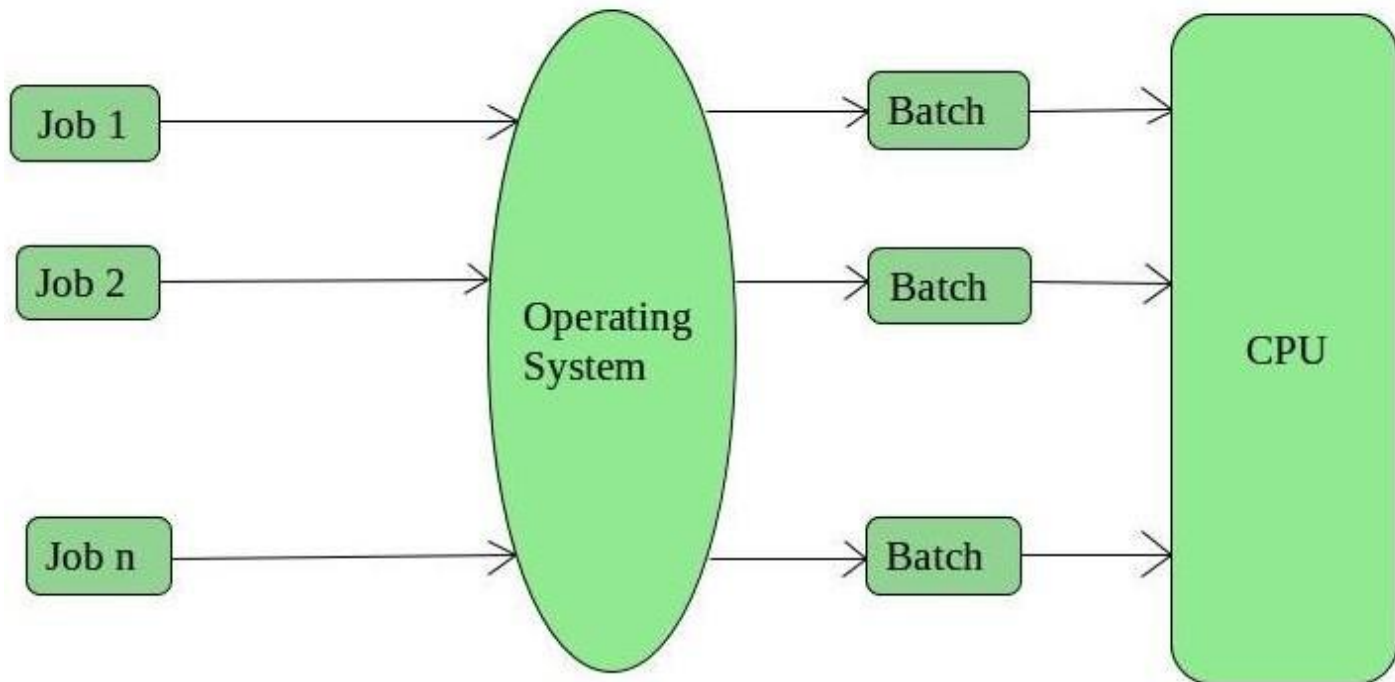
Character User Interface

- CUI stands for Character User Interface.
- This is a type of user interface where user interacts with computer using only keyboard. To perform any action a command is required.
- CUI is precursor of GUI and was used in most primitive computers.
- Most modern computers use GUI and not CUI.
Eg: MS-DOS has CUI.

Batch OS

- Batch processing operating system allows only one program to run at a time. These kinds of OS can still be found on some mainframe computer running batches of jobs.
- Batch processing OS works on a series of programs that are held in a queue.
- This type of operating system does not interact with the computer directly. There is an operator which takes similar jobs having the same requirement and group them into batches.
- Multiple users can share batch systems.
- Batch systems are hard to debug
- The other jobs will have to wait for an unknown time if any job fails
- Examples of Batch based Operating System: Payroll System, Bank Statements, etc.

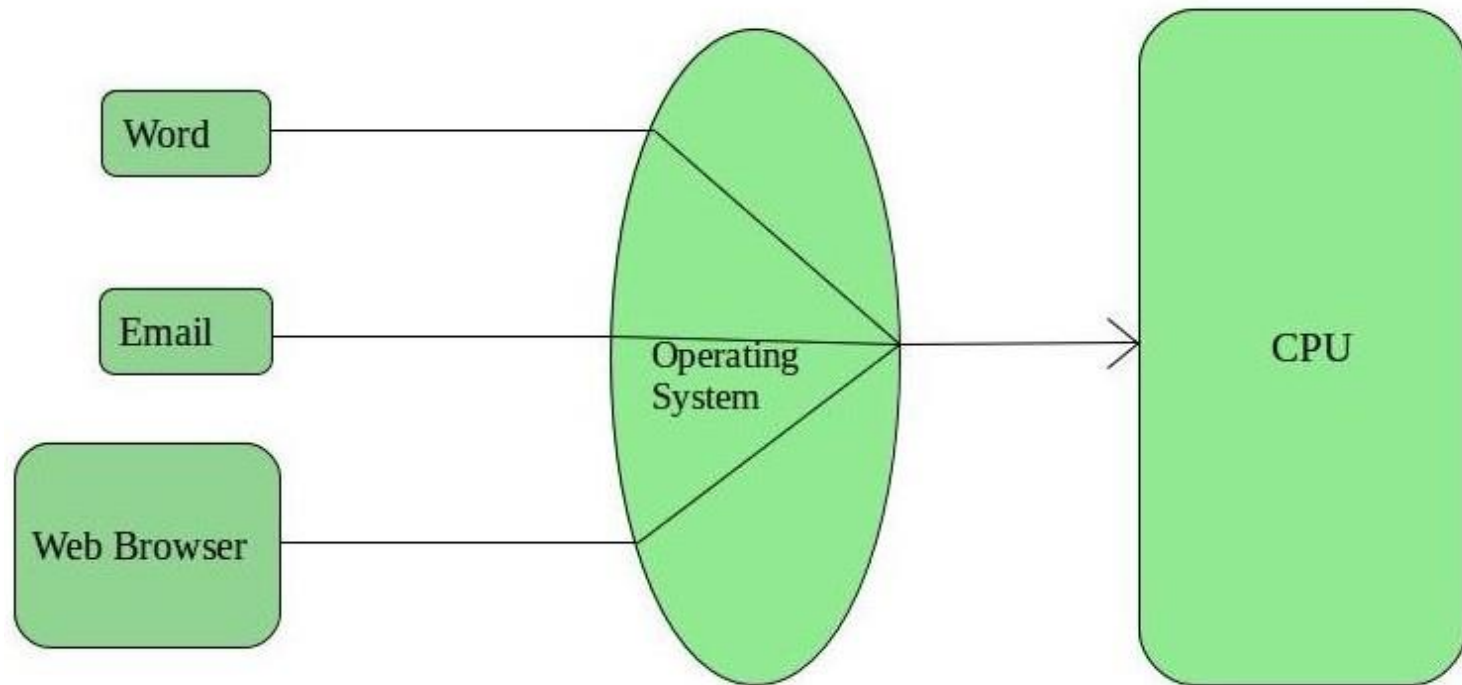
Batch OS



Multitasking OS

- The multitasking OS is also known as the time-sharing operating system as each task is given some time so that all the tasks work efficiently.
- This system provides access to a large number of users, and each user gets the time of CPU as they get in a single system. The tasks performed are given by a single user or by different users.
- The time allotted to execute one task is called a quantum, and as soon as the time to execute one task is completed, the system switches over to another task.
- Each task gets an equal opportunity & CPU idle time can be reduced
- One must have to take care of the security and integrity of user programs and data

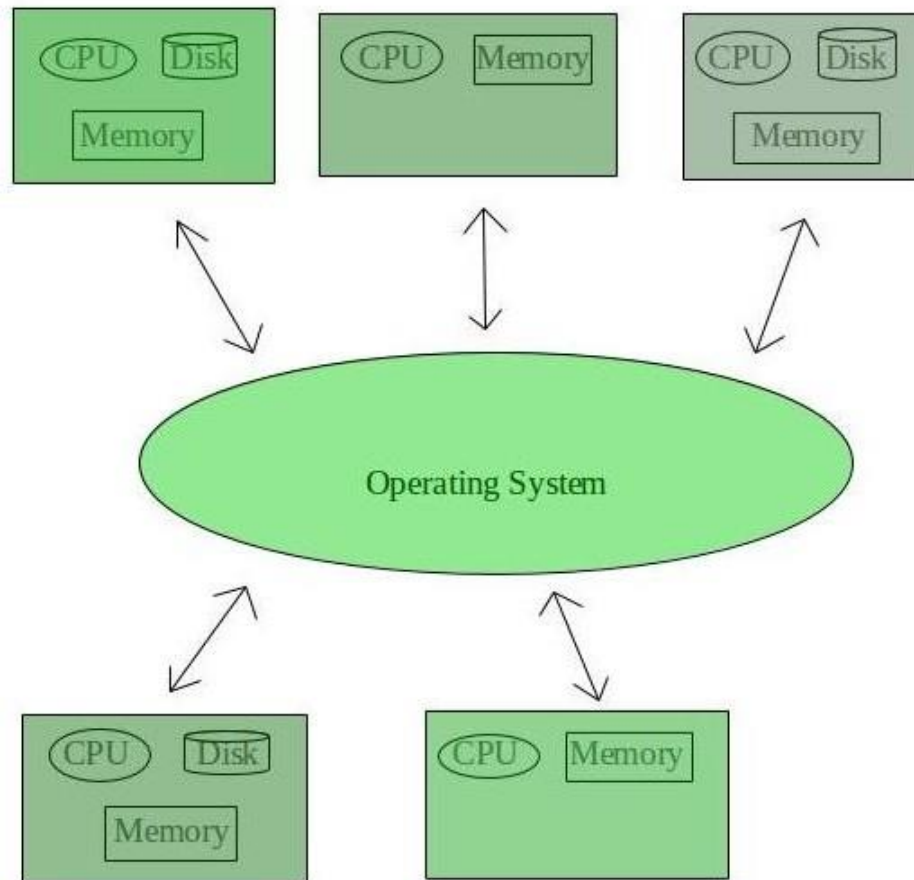
Multitasking OS



Distributed OS

- A distributed operating system is a recent advancement in the field of computer technology and is utilized all over the world that too with great pace.
- In a distributed OS, various computers are connected through a single communication channel. These independent computers have their memory unit and CPU and are known as loosely coupled systems. Failure of one system will not affect the other systems because all the computers are independent of each other.
- The system processes can be of different sizes and can perform different functions.
- The major benefit of such a type of operating system is that a user can access files that are not present on his system but another connected system. In addition, remote access is available to the systems connected to this network.
- Failure of the main network will stop the entire communication.
- Underlying software is complex.

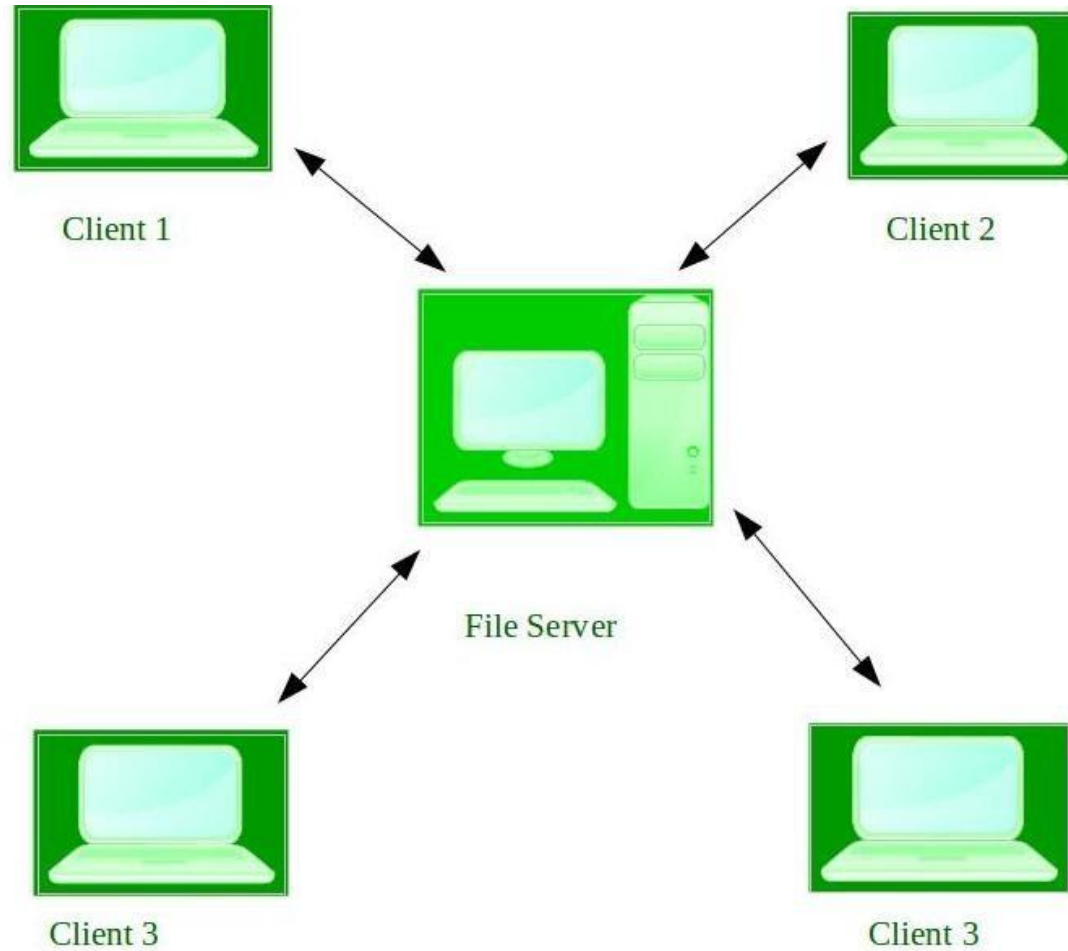
Distributed OS



Network OS

- These systems run on a server and provide the capability to manage data, users, groups, security, applications, and other networking functions.
- These types of operating systems allow shared access of files, printers, security, applications, and other networking functions over a small private network.
- One more important aspect of Network Operating Systems is that all the users are well aware of the underlying configuration, of all other users within the network, their individual connections, etc. and that's why these computers are popularly known as tightly coupled systems.
- Servers can be accessed remotely from different locations and systems.
- Server costs are high. Regular updates and maintenance are required.

Network OS



Real-Time OS

- These types of OSs serve real-time systems.
- The time interval required to process and respond to inputs is very small. This time interval is called response time.
- Real-time systems are used when there are time requirements that are very strict like missile systems, air traffic control systems, robots, etc.

Types of Real-Time OS

- **Hard Real-Time Systems:** These OSs are meant for applications where time constraints are very strict and even the shortest possible delay is not acceptable. These systems are built for saving life like automatic parachutes or airbags which are required to be readily available in case of any accident. Virtual memory is rarely found in these systems.
- **Soft Real-Time Systems:** The soft real-time OS is the operating system for applications where time constraint is not very strict. In a soft real-time system, an important task is prioritized over less important tasks, and this priority remains active until the completion of the task. Furthermore, a time limit is always set for a specific job, enabling short time delays for future tasks, which is acceptable. For Example, virtual reality, reservation systems, etc.

Open Source Operating System

- Open Source is a methodology or approach towards the design and development of software with the intention of giving the user the access to the code.
- If we use open source software, not only we will be able to use it but also be able to see how it works, debug it, modify it and redistribute it.
- Open source software is licensed in a way that marked it legal to use as many copies as user wants. There's a core difference between open source and free software, both represent the same core idea but the open resource allows the commercial utilization of code with the motive of profit.

Open Source Operating System

- An open source operating system is an operating system whose code has been made publicly and freely available to anyone who wants to see it and modify.
- Operating systems made available in source-code format rather than just binary closed-source
- Examples include GNU/Linux, BSD UNIX (including core of Mac OS X), and Sun Solaris

Open Source Operating System

- **Linux:** It is a popular operating system and it is free software as well as an open source software, i.e. Linux is freely available along with the underlying source code.
- Linux was originally written by Linus Torvalds, a student of Finnish University in 1991. After that, many programmers and students from all over the world have contributed a lot in its further refinement, development and improvement.
- Linux can be downloaded from www.linux.org.
- As an operating system, its aim is to provide interoperability (work together) with other operating systems and establish computing standards.